

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
import java.util.Scanner;

class DotException extends Exception {
    public DotException(String message) {
        super(message);
    }
}

class AtTheRateException extends Exception {
    public AtTheRateException(String message) {
        super(message);
    }
}

class DomainException extends Exception {
    public DomainException(String message) {
        super(message);
    }
}

public class Main {
    public static void validateEmail(String email) throws DotException,
    AtTheRateException, DomainException {
        int atCount = 0;
        for (char c : email.toCharArray()) {
            if (c == '@') atCount++;
        }
        if (atCount != 1 || email.startsWith("@") || email.endsWith("@")) {
            throw new AtTheRateException("Invalid @ usage");
        }

        int atIndex = email.indexOf('@');
```

```

String domainPart = email.substring(atIndex + 1);

int dotIndex = domainPart.lastIndexOf('.');
if (dotIndex == -1 || domainPart.startsWith(".") || domainPart.endsWith(".")) {
    throw new DotException("Invalid Dot usage");
}

if (domainPart.indexOf('.') != domainPart.lastIndexOf('.')) {
    throw new DotException("Invalid Dot usage");
}

String domain = domainPart.substring(dotIndex + 1);
if (!(domain.equals("in") || domain.equals("com") || domain.equals("net") ||
domain.equals("biz"))) {
    throw new DomainException("Invalid Domain");
}

for (int i = 1; i < email.length(); i++) {
    char prev = email.charAt(i - 1);
    char curr = email.charAt(i);
    if ((prev == '.' && curr == '.') || (prev == '@' && curr == '@')) {
        throw new DotException("Invalid Dot usage");
    }
}

System.out.println("Valid email address");
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String email = sc.nextLine();

    try {
        validateEmail(email);
    } catch (DotException e) {
        System.out.println("DotException: " + e.getMessage());
        System.out.println("Invalid email address");
    } catch (AtTheRateException e) {
        System.out.println("AtTheRateException: " + e.getMessage());
        System.out.println("Invalid email address");
    } catch (DomainException e) {
        System.out.println("DomainException: " + e.getMessage());
    }
}

```

```
}  
}  
} System.out.println("Invalid email address");  
}
```

Status : Correct

Marks : 10/10